

LLM-as-a-Discriminator

Can LLM Detect Synthetic Data?

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Privacy vs. Data Utility

The Tension

Organizations want to:

- ✓ Protect personal data
- ✓ Share useful insights

These goals often conflict

The Solution: Synthetic data

Generate artificial records that:

- ✓ Mimic real data patterns
- ✓ Protect individual privacy
- ✓ Enable data sharing

But: If someone can tell the data is synthetic, they can break privacy. Can they?

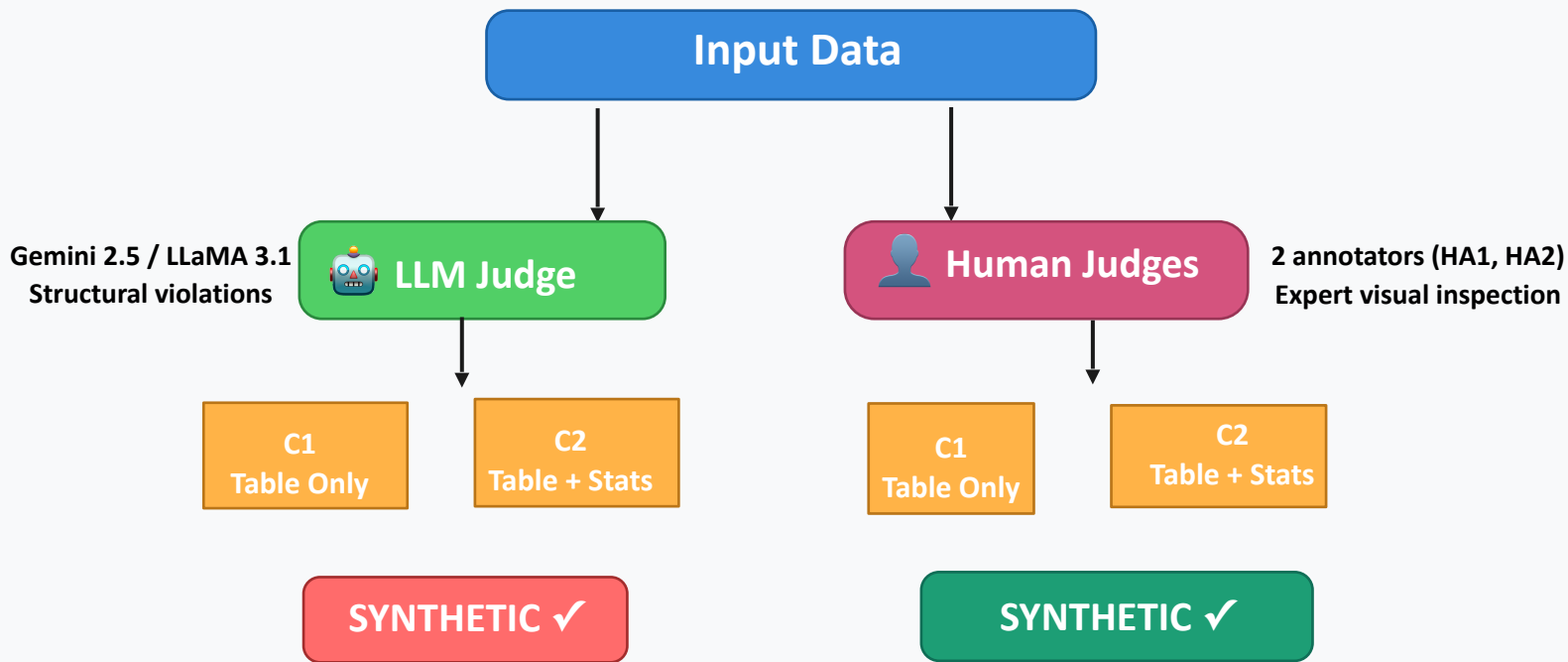
The question

Can a Large Language Model detect if the data is real or synthetic?

Why ask an LLM?

- No model access needed (black-box audit)
- Interpretable reasoning
- Scales easily

How we test: LLM vs Human Judges



Ideally: Both LLMs and Humans identify the same structural violations as the key discriminator

Experimental Design

451 valid experiments across:

Datasets: 2 census datasets: UCI Adult & ACS Census Income

Synthesis methods: 3 generators: CTGAN, TVAE, Gaussian Copula

Table 2. Trial counts per synthesizer and condition cell after discarding API failures.

Method	UCI Adult ACS Census			
	C1	C2	C1	C2
CTGAN	62	62	19	20
TVAE	57	52	19	19
Gaussian Copula	54	50	18	19
Total	173	164	56	58

LLM models: Gemini 2.5-Flash vs. LLaMA 3.1-8B

Threat conditions: C1 (table only) & C2 (table + metadata)

Baselines: Human annotators (2 people, 240 trials) + Logistic regression Classifier

Trials and failures: 3600 trials in total (3 synthesisers x 2 conditions x 200 trials per cell) : 90.6% failed (quota errors)

* Temperature is equal to 0 for reproducibility.

Code : <https://github.com/SlokomManel/LLM-as-a-Discriminator>

Two threat scenarios

Cond.	Discriminator input	Threat model
C1	20-row Markdown table + col-	Adversary with access to the data re-
	umn names + dataset shape	lease only
C2	C1 <i>plus</i> per-column statistics	Adversary who also has access to pub-
	(mean, std, quartiles, skew-	lished summary statistics
	ness, kurtosis), Shapiro–Wilk	
	results, top- <i>k</i> category fre-	
	quencies, Shannon entropy,	
	Pearson correlations	

Condition 1 (C1)

Table Only

LLM sees:

- 20 rows
- Column names
- Dataset shape

Adversary with data release only

Condition 2 (C2)

Table + Metadata

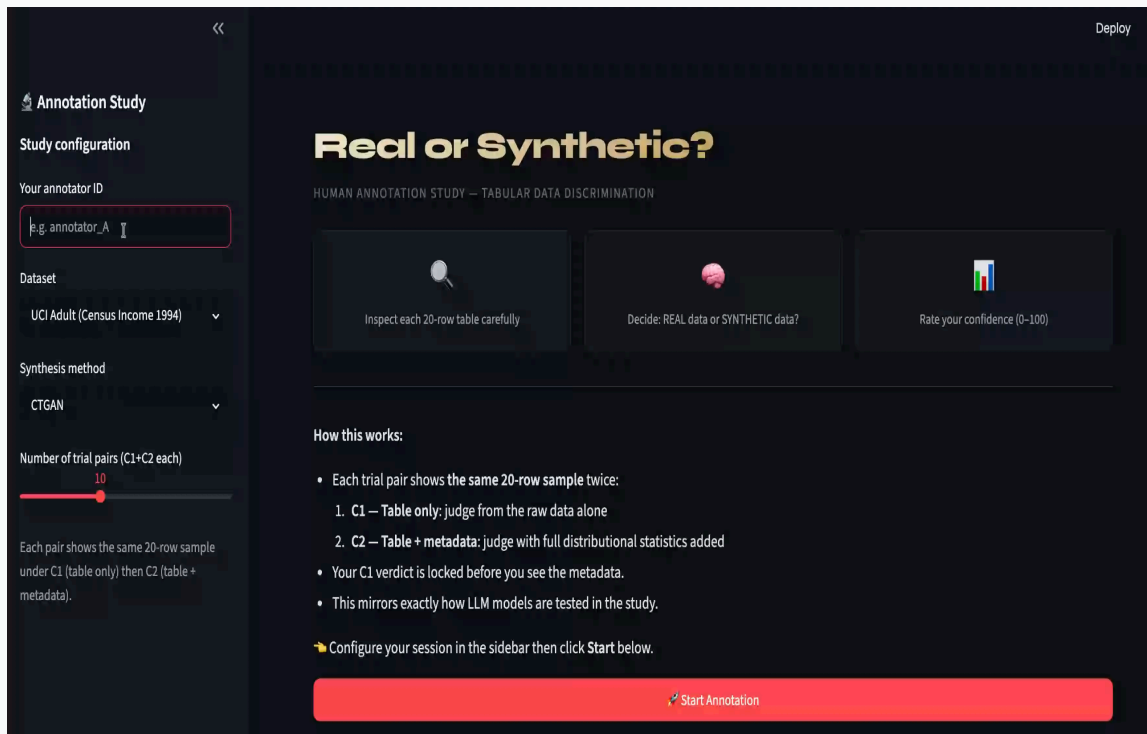
Same as C1 plus:

- Mean, std deviation
- Quartiles, skewness
- Category frequencies
- Correlations

Adversary with published stats

Human annotation study

- ✓ 2 human annotators
- ✓ 240 total trials (120 each)
- ✓ Same protocol as LLMs
- ✓ C1 then C2 per sample
- ✓ Real-time decision making
- ✓ Confidence scoring



Human annotator in action

Key Finding: Model matters more than method

UCI Adult results

LLaMA: DRS = 0% (labels everything as REAL)

Gemini: DRS = 100% for CTGAN & TVAE (perfect detection)

ACS Census results

LLaMA: DRS = 0% (labels everything as SYNTHETIC - opposite bias!)

Gemini: DRS = 100% under C1, varies under C2

The discriminator model makes as much difference as the synthesis method itself!

Metadata's surprising effect

Adding distributional metadata doesn't always help...

ACS Census with Gemini:

CTGAN:	C1: 100%	C2: 0%	↓ -100%
TVAE:	C1: 100%	C2: 25%	↓ -75%
Gaussian Copula:	C1: 100%	C2: 100%	→ stable

When synthetic marginals align with real data, metadata can confuse the detector!

What LLMs actually notice

Gemini identifies structural violations:

✗ Education Mapping

"Bachelors" → 13, "10th" → 6 (inconsistent)

✗ Impossible Combos

"Never-married" + "Husband" (4.5% of CTGAN samples)

**✗ Age/Education
Conflicts**

Age 18 with PhD (implausible)

**✗ Sparse Distribution
Anomalies**

Capital gains/losses filled in (90%+ zeros in real data)

What Humans and LLMs actually say

Structural violations: The primary clue

✗ Education mapping inconsistency

Gemini (LLM): *"'education_num' does not maintain a 1-to-1 mapping with 'education'; e.g., '10th' grade is assigned 4, not 6."*

✗ Marital Status & Relationship contradiction

Gemini (LLM): *"Marital status 'Never-married' paired with relationship type 'Husband' is logically impossible" (4.5% of CTGAN samples)*

Human Annotators: *"Never-married but also says Husband? That makes no sense."*

✗ Capital Gains/Loss Anomaly

LLaMA: *"High skewness in 'capital_gain' and 'capital_loss'; large number of zeros in capital_gain." (Real: 90%+ zeros. Synthetic: filled in.)*

✓ Both humans and LLMs independently identified the same structural violations as the key discriminator

How Humans compare to frontier LLMs

Human Annotators

Moderate Agreement

- Annotator 1: 66.4%
- Annotator 2: 77.9%

Reason: Spot structural violations independently

Frontier LLM (Gemini)

Matches or exceeds humans

- UCI Adult: DRS=100%
- ACS Census C1: DRS=100%
- Identifies same patterns
- Provides explicit reasoning
- Consistent across conditions

Practical Recommendations

LLM discrimination is a practical first-pass privacy audit tool (not a standalone guarantee)

When using LLM discrimination:

- ✓ Always report results per model (Gemini $\neq$ LLaMA)
- ✓ Report per condition (C1 and C2 may differ)
- ✓ Combine with formal baselines (record linkage, detection classifier, Membership Inference)
- ✓ Inspect LLM reasoning for explainability
- ✓ Consider data encoding effects (strings vs. integers)
- ✓ Don't rely on single metric for privacy guarantee

Limitations & Future Work

Study Constraints:

90.6% of planned trials failed due to API quotas (low per-cell N)

Only 8B and ~50B parameter models tested

Zero-shot prompting only (chain-of-thought unexplored)

Limited to 3 synthesis methods and 2 datasets

Diffusion & LLM-based generators not tested

Future Work:

→ Stable local deployments to avoid API sparsity

→ Broader synthesis methods (diffusion, LLM-based generators)

→ Larger human annotation studies

→ Joint privacy-utility reporting



The key insight

Model choice matters as much as synthesis method choice

When evaluating synthetic data privacy, always ask: Which LLM discriminator was used?

Questions?

Code & results:

github.com/SlokomManel/LLM-as-a-Discriminator

Paper available on arXiv